

Notes on morphological defectivity

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1 Introduction

- One curious fact about human language is that speakers can imagine, think about—even accidentally produce—words or sentences which they in some sense “can’t say”.

(1) Possible and impossible words of English (after Chomsky and Halle 1965):

- a. *brick* [bɹɪk]
- b. *blick* [blɪk]
- c. *bnick* [bnɪk]

(2) EB’s speech errors (from the Elikorpus):

There was a cake that there wasn’t eaten any of on the table.
I have people on my buddy list who I don’t know who are.
I wore clothes that I didn’t care whether got wet.
I’m sending you a sentence I’m trying to figure out why is hard.

- Let us call this pre-theoretic property of language **ineffability**.
- To be used for theorizing, observations about ineffability need to be related to expectations of what the grammars in question *ought* to generate.

1.1 Some English ineffability

- Any sensible grammatical description of English will have procedures yielding:
 - plurals of nouns
 - preterites and past participles of (lexical) verbs

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etc., and as such, the grammar identifies noun singulars and plurals, preterites, past participles, etc. as equivalence classes.

- For instance, some theories of word formation take the noun plural to be derived from the singular, creating an expectation that if one exists, the other ought to as well.
- Yet, *news* is strictly singular (**the news are bad*, **the new is bad*) and *pants* is strictly plural. Such words are known as **singularia tanta** and **pluralia tanta**, respectively.¹
- Thus, the behavior of *news* and *pants* violate our (naïve) expectations about noun inflection; *pant* is ineffable even though one might have expected otherwise.

(3) “Weather” verbs in English:

It is raining.
 ?She drizzles.
 ?I snowed.

(4) Defective irregular verbs in English (author’s judgments):

	bare	preterite		past participle	
a.	forgo	*forwent	*forgoed	forgone	(cf. Embick and Marantz 2008)
b.	stride	strode		*stridden *strided	(cf. Pullum and Wilson 1977:770)

(5) “Imperative-dominant” verbs in English:

- Beware! Go away! Scram! Shoo!**
 - Caesar was told to **beware** the Ides of March.
 The shopkeeper couldn’t wait for them to **go away**.
 Roaches started to **scram** when I turned the lights on.
 She shouted for the pigeons to **shoo**.
 - ?**Scramming** would be a good idea right about now.
 - *Roaches started **scramming** when I turned the lights on.
 - *I saw the roaches **scramming** when I turned the lights on.
 - *Roaches **scrammed** when I turned the lights on.
- For certain English verbs to be restricted in this fashion violates shared assumptions about grammatical modularity.

On most views of how the components of language interact, this is getting things backwards. Morphology ought simply to be the handmaid of grammatical meaning, producing forms where other components require them. By the principles of Phonology-Free Syntax and Morphology-Free Syntax, syntax should operate

¹I note the existence of affected *pant*, *trouser*, etc., in fashion writing (Garner 2016:669). Such nouns trigger singular verb agreement (e.g., *the pant is...*) in subject position, suggesting they are syntactically singular.

	inflection	derivation
category-changing?	rarely	often
“paradigmatic”?	yes	no
productive?	yes	maybe
semantically consistent?	usually	maybe
syntactic restrictions?	yes	no
position	usually further from stem	usually close to stem
portmanteau forms?	often	rarely
repeatable?	no	maybe

Table 1: Canonical properties of inflection vs. derivation (after Kroeger 2005:253, *cum grano salis*).

independent of the properties of individual word forms. Defectiveness represents an unwanted intrusion of morphological idiosyncrasy into syntax. From a morphological-internal perspective it is also problematic, since any coherent model of inflectional morphology assumes there will be productive, default mechanisms that permit the generation of a paradigm from any item whatsoever. ...**clearly it has been ill incorporated into our theories of language.** (Baerman and Corbett 2010:2)

1.2 Defining defectivity

- We are now in a position to discern a particular type of ineffability I call **defectivity**.
- First, I assume a distinction between **inflection** and **derivation**. This distinction is often fuzzy and difficult to apply, but most theories of word formation do draw it in some form; canonical diagnostic properties are listed in Table 1.

(6) Defectivity: A lexeme *L* is defective if and only if one or more of its possible inflectional variants is ineffable.

- Like any good definition, this notion of defectivity can be refined by comparing it to superficially similar phenomena. Let us consider a few such cases.

1.2.1 Analysis vs. synthesis

- As is well-known, Ancient Greek has synthetic perfect passives.

(7) ἐκεῖνοι κατ’ ἀξίαν ἐτιμήθησαν (Aeschin. 3.118)
 ‘Those men were deservedly honored’

- In Latin, however, the perfect passive can only be formed analytically, from a perfect passive participle and a copula.

- (8) maximē tamen **laudātus est** Mentor, dē quō suprā diximus. (Plin., *HN* 33.59)
'Mentor, however, who we spoke of above, was the most honored.'

- Since this is true of all Latin verbs, the sensible thing to say is:
 - that the grammar of Latin simply does not build synthetic perfect passives,²
 - that one has no reasonable expectation that it would,
 - and thus there is no sense in which a “possible” inflectional variant is ineffable.
- English can form comparative and superlative adjectives synthetically, by the addition of *-er/-est* suffixes (e.g., *smarter*, *tallest*) or analytically (e.g., *more intelligent*, *most Herculean*).³
- The distribution of these alternatives is complex but appears to be conditioned primarily by the metrical properties of the adjective stem (e.g., Hilpert 2008).
- Should one then regard the ill-formedness of **intelligenter* as an instance of defectivity?
- For Poser (1992) and Bresnan (2001), who claim phrases and lexical items are in grammatical “competition”, the answer is no: *more intelligent* is simply the correct comparative.
- Or to put it another way, under Poser’s proposal, the grammar merely “expects” that adjectives have comparative and superlative forms, but has no expectation as to whether they are analytic or synthetic.
- In contrast, Embick and Marantz (2008) reject the idea that word and phrases are in competition with one another. Rather, forms like **intelligenter* are not produced in the first place because of phonological conditions on *-er/-est* suffixation.
- Thus while Poser, Bresnan, and Embick and Marantz propose radically different analyses, they agree that...

...it is a mistake to group forms like **intelligenter* with forms like **forwent* and **forgoed*. (Embick and Marantz 2008:35)

1.2.2 Heteroclisis

- **Heteroclitics** are lexemes whose “paradigms” exhibit total stem suppletion, as in English *go/went* or *person/people*.
- If one assumes that *people* is simply the suppletive plural form of *person*—Kiparsky (1994:19) calls this **inherent morpholexical selection**—then there is no defectivity to speak of.

²Kiparsky (2005) and Kastner and Zu (2017) propose rather different grammatical explanations for Latin’s lack of synthetic perfect passives.

³I note that adjectives which form synthetic comparatives also permit analytic comparatives when used as a metalinguistic comparison; e.g., *he’s more tall than attractive*. Presumably this construction involves a different syntactic structure than ordinary comparatives.

- However, an alternative analysis would treat this as a case of (“accidental”) synonymy between two defective lexemes: singulare tantum *person* and plurale tantum *people*.⁴

1.2.3 Deponency

- In Latin, there are several hundred **deponents**, agentive verbs which take the form of passives—i.e., the tense, aspect, and agreement suffixes of passives—but which are semantically active.
- For example, compare partial paradigms of the ordinary *amō* ‘I love’ (9ab) and the deponent *hortor* ‘I exhort’ (9c).⁵

(9) Latin deponency (after Embick 2000):

a. present active:

am-ō	‘I love’	am-ā-mus	‘we love’
am-ā-s	‘you (sg.) love’	am-ā-tis	‘you (pl.) love’
am-a-t	‘s/he loves’	am-a-nt	‘they love’

b. present passive:

am- or	‘I am loved’	am-ā- mur	‘we are loved’
am-ā- ris	‘you (sg.) are loved’	am-ā- mini	‘you (pl.) are loved’
am-ā- tur	‘s/he is loved’	am-a- ntur	‘they are loved’

c. deponent, present active:

hort- or	‘I exhort’	hort-ā- ris	‘you (sg.) exhort’
hort-ā- tur	‘s/he exhorts’	hort-ā- mini	‘you (pl.) exhort’
hort-ā- mur	‘we exhort’	hort-a- ntur	‘they exhort’

- Depending on one’s analysis, one might consider the inability of these verbs to passivize to be a form of defectivity.⁶

1.2.4 Indeclinability

- Some words fail to exhibit characteristic inflectional variation, but can still be used in all expected syntactic contexts. In traditional grammar, adjectives and nouns which show this behavior are termed **indeclinable**.
- For example, Russian has several dozen indeclinable nouns, including acronyms (10a), loanwords (10b), and toponyms (10c).

⁴I note the existence of the pedantic plural *persons* (Garner 2016:314) in legal writing, and the similarly-elevated plural *peoples* used to refer to a plurality of ethnicities (*ibid.*, p. 779).

⁵I also note the existence of **semi-deponents**, Latin verbs which are deponent only in the perfect; see Embick 2000:191, fn. 12 for discussion.

⁶Grestenberger (2018) argues that passivization of deponents is possible in languages with richer voice morphology than found in Latin.

(10) Russian indeclinable nouns (Wade 2020):

- a. ВИЧ ‘HIV’, СССР ‘USSR’, МГУ ‘Moscow State University’, НЭП ‘NEP’ (W:\$40)
- b. авеню ‘avenue’, кашне ‘scarf’, резюме ‘résumé’, хиппи ‘hippie’ (W:\$36)
- c. Сочи ‘Sochi’, Токио ‘Tokyo’, Глазго ‘Glasgow’, Эри ‘Lake Erie’ (W:\$37)

- In the case of (10bc), these nouns end in vowels not found stem-finally in native vocabulary.
- However, indeclinable nouns can be found in all cases, showing that they are not defective.

(11) Indeclinable noun *радио* ‘radio’ in the SynTagRus corpus (Boguslavsky et al. 2009):

- a. **Радио** совмещает слабости печатных масс-медиа и телевидения... (nom.)
‘Radio combines the weaknesses of print media and television...’
- b. Ни телевидения, ни **радио** тогда еще не существовало. (gen.)
‘Neither television or radio existed then.’
- c. ...наука, рассказанная по **радио**, может быть интересна. (dat.)
‘...science told on radio could be interesting.’
- d. **Радио** она не слушала, газет не читала... (acc.)
‘She didn’t listen to the radio, she didn’t read newspapers...’
- e. ...автомобиль с кожаным салоном, **радио** и... (ins.)
‘...a car with a leather interior, a radio, and...’
- f. ...целых 3 минуты на **радио**, целых 5 минут на телевидении... (pre.)
‘...as much as three minutes on radio, five minutes on television...’

1.2.5 Taboo avoidance

- One might posit that taboo constructions (e.g., Tefera 1987, Treis 2005) are generated by the grammar, even if they are not produced in practice. If one recognizes a distinction between i-language and e-language, then taboo avoidance is not defectivity per se.

1.3 Three problems for the theory of defectivity

- I discern three major questions which any theory of defectivity will have to grapple with.

(12) Three problems for the theory of defectivity:

Actuation: how does defectivity emerge across time?

Encoding: how is defectivity grammatically encoded?

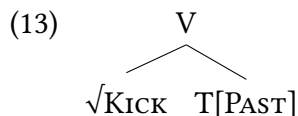
Acquisition: how is defectivity acquired by children?

- I now describe what is known—or can be conjectured—about these questions.

2 The encoding problem

2.1 English *forwent

- As noted in (4a), many English speakers cannot produce a preterite for $\sqrt{\text{FORGO}}$ (e.g., *forwent, *forgoed) though other inflected forms of this verb are unremarkable.
- Consider how this “gap in the paradigm” might be encoded grammatically.
- For concreteness, let us assume a **piece-based, realizational** theory of morphology:
 - First, ordinary syntactic operations build complex words.
 - Then, the morphology “realizes” a word by mapping the syntactic elements onto phonological pieces (called **morphs** or **vocabulary items**).
 - Realization rules may be conditioned to only apply in the presence of certain roots; there may also be an “elsewhere” (or “default”) rule.⁷
- These assumptions correspond roughly to the architecture of Distributed Morphology (Halle and Marantz 1993), henceforth DM, though that theory makes further assumptions not germane to the issue at hand.
- Let us assume that the structure of a past tense verb like *kicked* is roughly the following:



where T[past] denotes the past tense head.

(14) English past tense realization (after Embick and Marantz 2008:5):

$$\begin{array}{rclcl}
 & -t & / & \{\sqrt{\text{BEND}}, \sqrt{\text{LEAVE}}, \dots\} & \text{—} \\
 \text{T[PAST]} & \iff & -\emptyset & / & \{\sqrt{\text{HIT}}, \sqrt{\text{QUIT}}, \dots\} \quad \text{—} \\
 & -d & / & (\text{elsewhere}) &
 \end{array}$$

- In (14), /-d/ is the default or “elsewhere” case. This naturally expresses the fact that it is the past tense allomorph used for virtually all new verbs (e.g., *googled*, *hamburgled*).
- The problem is: (14) falsely predicts that the preterite of *forgo* is *forgoed. It is not at all clear how to modify these rules to correct this error without rejecting the idea that /-d/ is the “elsewhere” case.

⁷This recalls the Elsewhere Condition, an idea dating back to Pāṇini and later adopted in various forms by generative grammarians.

2.2 Spanish metaphony

- Many Spanish verbs with a mid *e* or *o* vowel in the final syllable in the stem in the infinitive—henceforth, mid stems—“diphthongize” (from *e* to *ie* [je] and *o* to *ue* [we], respectively) or “raise” (from *e* to *i*) in certain cells of their inflectional paradigm.

(15) Spanish present indicative metaphony:

	1sg.	2sg.	3sg.	1pl.	2pl.	3pl.	
a.	pego	pegas	pega	pegamos	pegáis	pegan	‘stick onto’
b.	niego	niegas	niega	negamos	negáis	niegan	‘deny’
c.	agosto	agostas	agosta	agostamos	agostáis	agostan	‘dry out’
d.	acuesto	acuestas	acuesta	acostamos	acostáis	acuestan	‘put to bed’
e.	elido	elides	elide	elidimos	elidís	eliden	‘elide’
f.	expido	expides	expide	expedimos	expedís	expiden	‘publish’

- It is largely unpredictable whether a mid stem will participate in these alternations.

	a.	pegar	pego	‘stick to’	b.	plegar	pliego	‘fold’
(16)	c.	cortar	corto	‘cut’	d.	contar	cuento	‘count’
	e.	divergir	diverjo	‘diverge’	f.	servir	sirvo	‘serve’

- There are a number of verbs—all 3rd conjugation mid stems like *agredir* ‘to attack’ and *aguerrir* ‘to harden’—which are defective in just those paradigm cells which would require the speaker to know whether the stem diphthongizes or raises (Maiden and O’Neill 2010).
- Again, it is not clear how to encode this defectivity. Just focusing on diphthongization:
 - Bybee and Pardo (1981) and Albright et al. (2001) argue diphthongization is produced by ordinary phonological rules; this provides no mechanism to trigger defectivity.
 - Harris (1969) uses a lexical diacritic [+D] to trigger diphthongization; it is unclear how either presence of absence of this diacritic would trigger defectivity.
 - Harris (1985) instead makes use of abstract URs to trigger diphthongization and raising; this too provides no mechanism to trigger defectivity.
 - Bermúdez-Otero (2013) analyzes diphthongization and raising as full suppletion. This could only be made to trigger defectivity if the stems—whatever their form—selected in diphthongization contexts are marked [–LEXICAL INSERTION] (cf. Halle 1973).

2.3 Polish masculine inanimate genitives

- The vast majority of masculine animates in Polish have a genitive singular (gen.sg.) in *-a* (e.g., *bażanta* ‘pheasant’, *kopacza* ‘digger’).

- In contrast, the majority of masculine inanimates (henceforth, masc.inan.) form a gen.sg. in *-u* (e.g., *leżaku* ‘deck chair’, *słoiku* ‘jar’), and *-a* is the minority pattern (e.g., *śmiecia* ‘trash’, *wiatraka* ‘windmill’).

(17) Counts of Polish masc.inan. gen.sg. allomorphs in UniMorph (McCarthy et al. 2020):

-a	-u	(other)
765	1,457	3

- Polish philologists (e.g., Westfal 1956, Dąbrowska 2001, Maunsch 2003) reluctantly accept that it is impossible to predict which allomorph a given masc.inan. noun will select.
- When a Polish speaker encounters certain rare or novel masc.inan. nouns, they are often unable or unwilling to use them in the gen.sg. (Kottum 1981, Gorman and Yang 2019): e.g., *drut* ‘wire’, *karabin* ‘rifle’, and *Göteborg* ‘Gothenburg’ are supposedly defective in the gen.sg.
- One could set up the realization rules so that the majority *-u* is the elsewhere case, along the lines of (14):

(18) Polish masc.inan. gen.sg. realization (to be revised):

$$[+GEN, -PL] \iff \begin{array}{l} /-a/ \quad / \quad \{\sqrt{\text{ŚMIEĆ}}, \sqrt{\text{WIATRAK}}, \dots\} \quad \text{—} \\ /-u/ \quad / \quad (\text{elsewhere}) \end{array}$$

- However, this falsely predicts that there should be no defectivity: any word not “listed” to select *-a* will select *-u*. One may therefore wish to revise these rules as follows.

(19) Polish masc.inan. gen.sg. realization (final version):

$$[+GEN, -PL] \iff \begin{array}{l} /-a/ \quad / \quad \{\sqrt{\text{ŚMIEĆ}}, \sqrt{\text{WIATRAK}}, \dots\} \quad \text{—} \\ /-u/ \quad / \quad \{\sqrt{\text{LEŻAK}}, \sqrt{\text{SŁOIK}}, \dots\} \quad \text{—} \end{array}$$

- Under standard assumptions, if a root is not on either of the two lists, the derivation will crash, deriving the observed defectivity.⁸

2.4 Tagalog actor focus

- Orgun and Sprouse (1999, 2009), henceforth OS, citing Schachter and Otnes (1972), claim that in Tagalog, “*-um-* does not occur with bases beginning with /m/ or /w/” (p. 292).

⁸Formally, in DM, this crash occurs because of the failure to realize uninterpretable case/number features.

(20) Tagalog *-um-* infixation (Orgun and Sprouse 2009):

a.	keri	k <u>u</u> meri	‘to carry’	(OS:30a)
	sulat	s <u>u</u> mulat	‘to write’	(OS:23)
b.	mahal	* <u>m</u> umahal	‘to become expensive’	(OS:28b)
	mura	* <u>m</u> umura	‘to become cheap’	(OS:28b)
	swer	*sw <u>u</u> mer	‘to swear’	(OS:29)

- OS formalize this using a variant of Optimality Theory (Prince and Smolensky 1993).
- They propose that the winning candidate selected by EVAL must then submitted to a language-specific CONTROL evaluation, which consists solely of **inviolable** markedness constraints. Any winning candidate which violates a CONTROL candidate is rendered defective.
 - Let $\gamma \subseteq U \times S$ be a function which produces (candidate) surface representations $S = s_1, s_2, \dots$ from underlying representations $U = u_1, u_2, \dots$. Then, γ is simply the composition of GEN and EVAL in the OT sense.
 - Let $\chi \subseteq S \times \{0, 1\}$ be a function from surface representations to their wellformedness such that a surface representation $s \in S$ is well-formed if and only if $\chi(s) = 1$. Then χ represents the CONTROL component.
 - Then, if for some underlying representation $u \in U$, let $s = \gamma(u)$. Then, if $\chi(s) = 0$, then u —or, if you prefer, the pair s, u —is defective.
- There are some major problems for OS’s claim, however.
- Many other Tagalog stems also fail to select an *-um-* infix but select another AF affix. For instance, Wiktionary lists *magmahal* and *magmura*.⁹
- This suggests that we are not dealing with defectivity since there is no reasonable expectation that a given stem will form an AF in *-um-* in the first place.
- OS further note that [...mum...] does occur:
 - there are some faithfully surfacing underlying forms containing /...mum.../, and
 - various reduplication processes may produce [...mum...].
- Given this, there *might* be a phonotactic restriction on *-um-* infixation (cf. the metrical restrictions on English synthetic comparatives), but whatever accounts for the unacceptability of **mumahal* is not purely phonotactic.

⁹As I understand it, the *mag-* AF prefix is causative, seemingly appropriate for the intended reading.

3 The acquisition problem

(This section is derived from Gorman and Yang 2019.)

- In his *Prolegomena to a theory of word formation* Halle (1973), henceforth H, proposes two accounts of defectivity, which he regards as equivalent.
- Under the first—and better-known—proposal, words like **forwent* are generated by the word formation component of the grammar, but are marked with a feature $-\text{LEXICAL INSERTION}$ preventing their actual use.
- This idea is implicit in accounts of lexical gaps (e.g., Pertsova 2005, 2016) employing the notion of **lexical conservatism** (Steriade 1998, 2008).

(21) Lexical conservatism: Learners require explicit positive evidence before using a novel allomorph.

- However, this account seems difficult to reconcile with the unbounded creativity of word formation: children do pass the *wug*-test (Berko 1958) after all.
- As an alternative H (fn. 1) suggests that only those items undergoing “nonproductive” word formation rules can bear the feature $-\text{LEXICAL INSERTION}$, with all other forms marked $+\text{LEXICAL INSERTION}$ by default.
- Hetzron (1975), arguing against H’s first proposal, seems to endorse H’s second proposal linking defectivity to unproductivity, as do Albright (2003, 2010), Anderson (2010), Hudson (2000), Löwenadler (2010), and Maiden and O’Neill (2010), among others.

(22) Two questions for acquiring defectivity (after H):

- a. Is the word formation rule productive?
- b. Is the form marked $-\text{LEXICAL INSERTION}$?

- Gorman and Yang (2019) propose children answer (22a) by applying the Tolerance Principle.

(23) Tolerance Principle (Yang 2005, 2016):

Suppose a rule R is applicable to N items in a learner’s vocabulary, of which e are exceptions that do not follow R . Then, R is productive if and only if $e \leq \theta_N$ where

$$\theta_N := \frac{N}{\ln N}.$$

(24) Condition on lexical gaps (Yang 2016:142):

Let there be S rules each affecting N_i lexical items ($1 \leq i \leq S$), and let $N = \sum_i N_i$. Gaps arise if and only if

$$\forall i. \sum_{j \neq i} N_j < \theta_N.$$

- Gorman and Yang’s case studies include English irregular verbs (4), Spanish metaphony (§2.2), Polish masc.inan. genitives (§2.3), and Russian 2nd conjugation verbs (§4, see below).
- When the answer to (22a) is negative (i.e., the WFR is non-productive), it is not difficult for a child to discover that a wordform bears the non-default +LEXICAL INSERTION feature.
- For example, the Tolerance Principle predicts that both *-a* and *-u* are unproductive exponents of the masc.inan. gen.sg..

(25) Polish masc.inan. gen.sg. Tolerance computations ($R = 2, 222$):

- a. *-a*: $e = 765 > \theta \approx 200$
- b. *-u*: $e = 1, 457 > \theta \approx 115$

- Then, a child who hears *słoiku* or *śmiecia*—and correctly parses them as the gen.sg. of *słóik* ‘jar’ or *śmieci* ‘trash’, respectively—may immediately conclude that *słoiku* or *śmiecia* are +LEXICAL INSERTION.

4 The actuation problem

- Even less is known about how defectivity arises in a grammar.
- There are several dozen Russian verbs in the 2nd conjugation (i.e., those with a *-e-* or *-i-* theme vowel) which are defective in the first person singular (1sg.) non-past.

(26) Russian 1sg. non-past verbs in /z/ (Halle 1973):

a.	возить	[vɐ.'zʲitʲ]	‘to transport’	вожу	[vɐ.'zu]	‘I transport’
	пронзить	[prɐn.'zʲitʲ]	‘to stab’	пронжу	[prɐn.'zu]	‘I stab’
b.	дерзить	[dʲɪr.'zʲitʲ]	‘to be rude’	*держу	[dʲɪr.'zu]	‘I am rude’
	елозить	[jɪ.'lɔ.zʲitʲ]	‘to crawl’	*еложу	[jɪ.'lɔ.zu]	‘I crawl’

(27) Other Russian defective verbs (Pertsova 2016):

/s, ʃ/:	колесить	‘to take a detour’	*коле <u>ш</u> у	‘I take a detour’
/z, zʲ/:	лазить	‘to climb’	*ла <u>ж</u> у	‘I climb’
/t, ʈ/:	ощутить	‘to sense’	*ощу <u>щ</u> у	‘I sense’
/d, zʲ/:	победить	‘to win’	*побе <u>ж</u> у *побе <u>жд</u> у	‘I win’

- The final dental—i.e., /s, z, t, d/—consonant of the stem undergoes retroflexion/palatalization in this context, but verbs with or without these mutations are ungrammatical, and speakers must resort to paraphrase (e.g., *я одержу победу*, lit. ‘I will gain victory’).
- Since virtually every other Russian verb has a 1sg. non-past form, this is a clear case of defectivity under the proposed definition.
- Baerman (2008), henceforth B, heroically traces the origins of defectivity in Russian, focusing on the mutation of /d/.
- B notes that there is variation in mutation in the pre-revolutionary period. For example, Tolstoy uses *убежу* ‘I convince’ with /d/ mutating to [ʒ], whereas his contemporary Dostoevsky favors *убежду* with the (now non-standard) mutation of /d/ to [ʒd].
- B surveys 13 pre-revolutionary dictionaries, and finds that most verbs that are now defective are listed as non-alternating—i.e., not undergoing mutation in the 1sg. non-past—before 1917.
- Thus one may posit that stigmatization of non-alternation in the 1sg. non-past somehow actuated defectivity in Russian.

One extreme would be to view defectiveness [is] the surface manifestation of underlying non-alternation...**defective verbs are really non-alternating verbs whose production is aborted by the superficial constraints of the literary standard.** (B:95)

- It is interesting to note that English neologistic verbs whose stems end in /s, z, t, d/ (e.g., *фрилансить* ‘to freelance’, *зафрендить* ‘to add as a friend on social media’) appear to have entered the language as defective (Pertsova 2016:24, Slioussar and Kholodilova 2011).

5 Caveats

- I conclude my remarks with two lingering caveats.

5.1 Interspeaker variation

- Defectivity seems to give rise to substantial—and poorly understood—interspeaker variation in acceptability. Two examples follow.
- In German, certain object-incorporated verbs cannot be placed in matrix clauses.

(28) German (in)separable verbs (Vikner 2005):

- a. Sie will **bausparen.** (V:8a)
 she wants to building-save
 ‘She wants to save with a building society.’

b. ***Spart** er **bau**? (V:8c)
saves he building

c. ***Bauspart** er? (V:8d)
building-saves he
Intended: ‘Does he preserve via building society?’

- Other German verbs showing these restrictions include *bauchlanden* ‘belly-flop’ (lit. ‘belly-land’), *generalüberholen* ‘give s.t. a general overhaul’, *kopfrechen* ‘do mental arithmetic’ (lit. ‘head-count’), and *staubsaugen* ‘vacuum’ (lit. ‘dust-suck’).
- However, Holmberg (1976:78, cited in den Dikken 2003) reports that just 20% of German speakers interviewed reject sentences like (28b).
- Ross (1996:15) reports that Tagalog *um*-infixation (§2.4) is permitted for English borrowings beginning with /Cm, Cw/ (e.g., for *twitter* ‘id.’, Ross gives [twumitər, tumwitər]), but OS claim such forms are unacceptable.

5.2 Modernity?

- Defectivity is not purely a product of modernity, attendant ideologies about “standard language” (in the sense of Lippi-Green 1997), or institutions (e.g., national language academies) enforcing those ideologies.
- As noted by Baerman and Corbett (2010:2), Latin grammars (e.g., Neue 1866:§115f., Allen and Greenough 1903:§103) list **monoptotes**, **diptotes**, and **triptotes**, nouns which can occur in at most two, or three of the six cases.¹⁰
- Defectivity is by no means limited to languages *mit an armey un flot* (M. Weinreich, possibly apocryphal). For example, in Chickasaw, a Muskogean language, certain verbs (e.g., ‘remember’) simply cannot be used with 1st- or 2nd-person objects (Munro 2005:125).

A quick plug

I have published a series of defectivity case studies on my blog:

<https://www.wellformedness.com/blog/category/language/defectivity/>

References

- Albright, Adam. 2003. A quantitative study of Spanish paradigm gaps. In *Proceedings of the 22th West Coast Conference on Formal Linguistics*, 1–14.
- Albright, Adam. 2010. Lexical and morphological conditioning of paradigm gaps. In *Defective Paradigms: Missing Forms and What They Tell Us*, ed. Matthew Baerman, Greville G. Corbett, and Dunstan Brown, 117–151. Oxford University Press.

¹⁰These in turn are known to us from the *testimonia* of ancient grammarians.

- Albright, Adam, Argelia Andrade, and Bruce Hayes. 2001. Segmental environments of Spanish diphthongization. *UCLA Working Papers in Linguistics* 7:117–151.
- Allen, Joseph H., and J. B. Greenough. 1903. *New Latin Grammar*. Ginn & Company.
- Anderson, Stephen R. 2010. Failing one's obligations: Defectiveness in rumantsch reflexes of DĒBĒRE. In *Defective Paradigms: Missing Forms and What They Tell Us*, ed. Matthew Baerman, Greville G. Corbett, and Dunstan Brown, 19–34. Oxford University Press.
- Baerman, Matthew. 2008. Historical observations on defectiveness: the first singular non-past. *Russian Linguistics* 32:81–97.
- Baerman, Matthew, and Greville G. Corbett. 2010. Defectiveness: typology and diachrony. In *Defective Paradigms: Missing Forms and What They Tell Us*, ed. Matthew Baerman, Greville G. Corbett, and Dunstan Brown, 19–34. Oxford University Press.
- Berko, Jean. 1958. The child's learning of english morphology. *Word* 14:150–177.
- Bermúdez-Otero, Ricardo. 2013. The Spanish lexicon stores stems with theme vowels, not roots with inflectional class features. *Probus* 25:3–103.
- Boguslavsky, Igor, Leonid Iomdin, Tatyana Frolova, and Svetlana Timoshenko. 2009. Development of a Russian tagged corpus with lexical and functional annotation. In *Metalanguage and Encoding Scheme Design for Digital Lexicography: MONDILEX Third Open Workshop*, 83–90.
- Bresnan, Joan. 2001. Explaining morphosyntactic competition. In *Handbook of Contemporary Syntactic Theory*, ed. Mark Baltin and Chris Collins, 1–44. Blackwell.
- Bybee, Joan, and Elly Pardo. 1981. Morphological and lexical conditioning of rules: experimental evidence from Spanish. *Linguistics* 19:937–968.
- Chomsky, Noam, and Morris Halle. 1965. Some controversial questions in phonological theory. *Journal of Linguistics* 1:97–138.
- den Dikken, Marcel. 2003. When particles won't part. Ms., Graduate Center, City University of New York.
- Dąbrowska, Ewa. 2001. Learning a morphological system without a default: the Polish genitive. *Journal of Child Language* 28:545–574.
- Embick, David. 2000. Features, syntax, and categories in the Latin perfect. *Linguistic Inquiry* 31:185–230.
- Embick, David, and Alec Marantz. 2008. Architecture and blocking. *Linguistic Inquiry* 39:1–53.
- Garner, Bryan A. 2016. *Garner's Modern Usage*. Oxford University Press, 4th edition.
- Gorman, Kyle, and Charles Yang. 2019. When nobody wins. In *Competition in Inflection and Word Formation*, ed. Wolfgang U. Dressler, Franz Rainer, Francesco Gardani, and Hans Christian Luschützky, 169–193. Springer.
- Grestenberger, Laura. 2018. Deponency in finite and nonfinite contexts. *Language* 94:487–526.
- Halle, Morris. 1973. Prolegomena to a theory of word formation. *Linguistic Inquiry* 4:3–16.
- Halle, Morris, and Alec Marantz. 1993. Distributed morphology and the pieces of inflection. In *The View from Building 20: Essays in Linguistics in Honor of Sylvain Bromberger*, ed. Kenneth Hale and Samuel Keyser, 111–176. MIT Press.
- Harris, James W. 1969. *Spanish Phonology*. MIT Press.

- Harris, James W. 1985. Spanish diphthongisation and stress: a paradox resolved. *Phonology* 2:31–45.
- Hetzron, Robert. 1975. Where the grammar fails. *Language* 51:859–872.
- Hilpert, Martin. 2008. The English comparative: language structure and language use. *English Language and Linguistics* 12:395–417.
- Holmberg, Märta Åsdahl. 1976. *Studien zu den verbalen Pseudokomposita im Deutschen*. Acta Universitatis Gothoburgensi.
- Hudson, Richard. 2000. *i amn't. *Language* 76:297–323.
- Kastner, Itamar, and Vera Zu. 2017. Blocking and paradigm gaps. *Morphology* 27:643–684.
- Kiparsky, Paul. 1994. Allomorphy or morphophonology? In *Trubetzkoy's Orphan*, ed. Rajendra Singh, 13–31. John Benjamins.
- Kiparsky, Paul. 2005. Blocking and periphrasis in inflectional paradigms. *Yearbook of Morphology* 113–135.
- Kottum, Steiner E. 1981. The genitive singular form of masculine nouns in Polish. *Scando-Slavica* 27:179–186.
- Kroeger, Paul. 2005. *Analyzing Grammar: An Introduction*. Cambridge University Press.
- Lippi-Green, Rosina. 1997. *English with an Accent: Language, Ideology, and Discrimination in the United States*. Routledge.
- Löwenadler, John. 2010. Restrictions on productivity: defectiveness in Swedish adjective paradigms. *Morphology* 20:70–107.
- Maiden, Martin, and Paul O'Neill. 2010. On morphomic defectiveness: evidence from the Romance languages of the Iberian peninsula. In Baerman et al. (2010), pages 103–124.
- Maunsch, Hanna. 2003. Current alternations in inflection of Polish masculine inanimate nouns in the singular: a pilot study. *Investigationes Linguisticae* 9:4–21.
- McCarthy, Arya D., Christo Kirov, Matteo Grella, Amrit Midhi, Patrick Xia, Kyle Gorman, Ekaterina Vylomova, Sebastian Mielke, Garrett Nicolai, Miikka Silfverberg, Stefanie Reed, Yuval Pinter, Cassandra L. Jacobs, Ryan Cotterell, Mans Hulden, and David Yarowsky. 2020. UniMorph 3.0: universal morphology. In *Proceedings of the 12th Language Resources and Evaluation Conference*, 3922–3931.
- Munro, Pamela. 2005. Chickasaw. In *Native Languages of the Southeastern United States*, ed. Heather K. Hardy and Janine Scancarelli, 114–156. University of Nebraska Press.
- Neue, Friedrich. 1866. *Formenlehre der lateinischen Sprache*. H. Lindemann.
- Orgun, C. Orhan, and Ronald Sprouse. 2009. Hard constraints in Optimality Theory. In *Modeling Ungrammaticality in Optimality Theory*, ed. Curt Rice and Sylvia Blaho, 97–116. Equinox.
- Orgun, Cemil Orhan, and Ronald L. Sprouse. 1999. From MPARSE to CONTROL: deriving ungrammaticality. *Phonology* 16:191–224.
- Pertsova, Katya. 2005. How lexical conservatism can lead to paradigm gaps. *UCLA Working Papers in Linguistics* 11:13–30.
- Pertsova, Katya. 2016. Transderivational relations and paradigm gaps in Russian verbs. *Glossa* 1:13.

- Poser, William J. 1992. Blocking of phrasal constructions by lexical items. In *Lexical Matters*, ed. Ivan A. Sag and Anna Szabolcsi, 111–130. CSLI.
- Prince, Alan, and Paul Smolensky. 1993. Optimality theory: Constraint interaction in generative grammar. Technical report TR-2, Rutgers University Center for Cognitive Science. Published by MIT Press, 2004.
- Pullum, Geoffrey K., and Deidre Wilson. 1977. Autonomous syntax and the analysis of auxiliaries. *Language* 53:741–788.
- Ross, Kie. 1996. Floating phonotactics: Variability in reduplication and infixation of Tagalog loanwords. Master's thesis, University of California, Los Angeles.
- Schachter, Paul, and Fe T. Otanes. 1972. *Tagalog Reference Grammar*. University of California Press.
- Slioussar, Natalia, and Maria Kholodilova. 2011. Paradigm leveling in non-standard Russian. In *Formal Approaches to Slavic Linguistics: The Second MIT Meeting 2011*, 243–258.
- Steriade, Donca. 1998. Lexical conservatism. In *Linguistics in the Morning Calm: Selected Papers from SICOL 1997*, 157–179.
- Steriade, Donca. 2008. A pseudo-cyclic effect in Romanian morphophonology. In *Inflectional Identity*, ed. Asaf Bachrach and Andrew Nevins, 313–360. Oxford University Press.
- Tefera, Anbessa. 1987. Ballišša: women's speech among the Sidama. *Journal of Ethiopian Studies* 20:44–59.
- Treis, Yvonne. 2005. Avoiding their names, avoiding their eyes: how Kambaata women respect their in-laws. *Anthropological Linguistics* 47:292–320.
- Vikner, Sten. 2005. Immobile complex verbs in Germanic. *Journal of Comparative Germanic Linguistics* 8:85–117.
- Wade, Terence. 2020. *A Comprehensive Russian Grammar*. Wiley Blackwell, 4th edition.
- Westfal, Stanisław. 1956. *A Study in Polish Morphology: the Genitive Singular Masculine*. Mouton.
- Yang, Charles. 2005. On productivity. *Linguistic Variation Yearbook* 5:333–370.
- Yang, Charles. 2016. *The Price of Linguistic Productivity: How Children Learn to Break the Rules of Language*. MIT Press.
- Zaliznyak, Andrey A. 1977. *Grammatičeskij slovar' russkogo jazyka*. Russkij jazyk.

A Defectivity in Russian nouns

(29) Russian nouns defective in the gen.pl. (Zaliznyak 1977):

- a. *Р.мн. нет* 'gen.pl. does not exist'

беремя	'burden' (n. inan.)
камка	'damask' (f. inan.)
майя	'lightweight cotton fabric' (f. inan.)
мгла, мга	'darkness; haze' (f. inan.)
мзда	'bribe' (f. inan.)
пенька	'hemp' (f. inan.)
полумгла, полутьма	'twilight' (f. inan.)
райя	'rayah' (f. inan.)
- b. *Р.мн. затрудн.* 'gen.pl. difficult'

ага	'agha' (f. anim.)
айва	'quince' (f. inan.)
баба-яга	'Baba Yaga' (f. anim.)
балда	'fool' (m./f. anim.)
башка	'noggin' (f. inan.)
брюзга	'grouch' (m./f. anim.)
гомоса	'fidgeter' (m./f. anim.)
дуда	'bagpipe' (f. inan.)
егоза	'fidgeter' (m./f. anim.)
зуда	'annoying person' (m./f. anim.)
иго	'yoke' (n. inan.)
кайма	'hem' (f. inan.)
карга	'hag' (f. anim.)
киса	'purse' (f. inan.)
корма	'stern, rear end' (f. inan.)
кочерга	'fire poker' (f. inan.)
краса	'beauty' (f. inan.)
кума	'godmother' (f. anim.)
лапта	'lapta' (f. inan.)
мечта	'dream' (f. inan.)
мольба	'prayer, request' (f. inan.)
мулла	'mullah' (m. anim.)
пакля	'oakum' (f. inan.)
раба	'female slave' (f. anim.)
радиоэхо	'radio echo' (n. inan.)
сайга	'saiga' (f. anim.)

B Defectivity in Russian verbs

(30) Russian verbs defective in 1sg. non-past. (Pertsova 2016):

басить	'to use a deep voice'	мерзить	'to be nasty'
бдеть	'to keep a watch'	мутить	'to roil'
бороздить	'to furrow'	нудить	'to be annoying'
бузить	'to fulminate'	обезлесить	'to deforest'
вызвездить	'to cover with stars'	обезлошадить	'to take away a horse'
галдеть	'to make a lot of noise'	обессмертить	'to immortalize'
(при)гвоздить	'to nail down'	обрушить	'to Russify'
грезить	'to (day)dream'	ощутиться	'to find oneself'
гудеть	'to buzz; to honk'	парусить	'to sail around'
дерзить	'to be rude'	победить	'to win'
дубасить	'to beat violently'	претить	'to arouse disgust'
дудеть	'to play a pipe'	приютить	'to shelter'
елозить	'to fidget'	пылесосить	'to vacuum'
ерундить	'to speak nonsense'	рысить	'to trot'
желтить	'to turn yellow'	сбондить	'to steal'
завязать	'to get sucked into s.t.'	сбрендить	'to go crazy'
застить	'to obstruct a view'	соседить	'to neighbor s.t.'
затмить	'to eclipse'	убедить	'to convince'
зудеть	'to itch'	угобзить	'to fertilize'
каверзить	'to do mischief'	угораздить	'to urge'
капризить	'to be capricious'	умилосердить	'to compel to mercy'
колесить	'to drive around'	форсить	'to swagger'
кудесить	'to do magic'	чадить	'to emit fumes'
кутить	'to carouse'	читать	'to revere'
лазить	'to climb'	чудесить	'to do magic'
лебезить	'to fawn'	чудить	'to act strangely'
лишить	'to be sly'	шелестеть	'to rustle'
лихорадить	'to have a fever'	шерстить	'to search vehemently'
ляамзить	'to steal, swipe'	шкодить	'to misbehave'
(с)ляпсить	'to steal, swipe'		